

Building a 24-Bit Ultrasonic Data Acquisition System

For Railroad Wheel Testing Applications

*A Complete Guide from Component Procurement
through Field Deployment*

System Specifications:

24-bit Resolution • 500 kSPS Sample Rate • 80 kHz Bandwidth
Field-Portable • Battery-Powered • Direct MATLAB Integration

Table of Contents

1. System Overview

This document provides a complete guide to building a field-deployable 24-bit ultrasonic data acquisition system for capturing acoustic emissions from railroad wheels. The system is designed around commercial evaluation hardware from Analog Devices and Digilent, requiring minimal custom development while achieving professional-grade measurement quality.

1.1 System Architecture

The complete signal chain flows from an ultrasonic microphone through an anti-aliasing filter to a precision 24-bit ADC, with digital data transferred via FPGA to a host laptop over galvanically-isolated Gigabit Ethernet. The key components work together as follows:

The Avisoft CM24/CPMA-5V microphone captures ultrasonic acoustic emissions in the 2 kHz to 200 kHz frequency range with approximately 500 mV/Pa sensitivity and very low self-noise of approximately 15 dB SPL. Its differential output of plus or minus 5V directly matches the AD4630-24 ADC input range when using a 5V reference, eliminating the need for signal conditioning beyond anti-aliasing filtering.

The EVAL-AD4630-24FMCZ evaluation board provides 24-bit resolution at sample rates up to 2 MSPS. For the target 80 kHz signal bandwidth, a 500 kSPS sample rate provides adequate margin while keeping data rates manageable. The evaluation board connects directly to the ZedBoard via the FMC-LPC connector.

The Digilent ZedBoard hosts a Xilinx Zynq-7020 SoC combining FPGA programmable logic with dual ARM Cortex-A9 processors. Analog Devices provides pre-built HDL reference designs, Linux drivers, and Python libraries that eliminate custom FPGA development entirely.

1.2 Key Design Decisions

Several architectural choices optimize this system for field deployment and measurement quality:

Gigabit Ethernet interface: **Strongly preferred over USB** because standard Ethernet provides inherent galvanic isolation through transformer magnetics in the PHY. This eliminates ground loops that would otherwise inject 50/60 Hz hum and switching noise into precision measurements. USB directly connects ground planes and requires additional isolation hardware.

LiFePO4 battery power: **Recommended over USB-C Power Delivery or AC adapters** for field use. LiFePO4 chemistry provides superior voltage stability throughout discharge, maintaining 12-13V for over 90 percent of capacity. Standard Li-ion packs drop to 9V when depleted, potentially causing system instability.

Separate preamp power: **The Avisoft sensor requires isolated 5V power** rather than drawing from ZedBoard's USB ports. The ZedBoard's digital switching noise from DDR3, Ethernet PHY, and Zynq power planes would degrade ultrasonic measurement quality.

Capture-then-transfer workflow: **For 10-second burst captures**, buffering data in ZedBoard's 512 MB DDR3 before bulk transfer to MATLAB provides guaranteed sample continuity without real-time network timing constraints.

1.3 Performance Specifications

Parameter	Specification	Notes
ADC Resolution	24 bits	True resolution, not effective bits
Sample Rate	500 kSPS (configurable to 2 MSPS)	500 kSPS adequate for 80 kHz BW
Signal Bandwidth	2 kHz to 80 kHz	Limited by anti-aliasing filter
Input Range	$\pm 5V$ differential	Matches Avisoft preamp output
Channels	2 simultaneous	Single channel sufficient for most testing
Data Interface	Gigabit Ethernet	80-100 MB/s practical throughput
Buffer Capacity	512 MB DDR3	Supports 170+ seconds at 500 kSPS
Battery Runtime	10+ hours	With 153Wh LiFePO4 pack

2. Component Procurement

All components are available from standard electronics distributors. The complete bill of materials totals approximately \$1,500 to \$2,000 excluding the Avisoft microphone. This section provides specific part numbers and supplier recommendations.

2.1 Core System Components

Item	Description	Part Number	Supplier	Price
Microphone	Avisoft CM24/CMPA-5V with 200V polarization	Custom order CM24-5V	Avisoft Bioacoustics	€600-800
ADC Board	EVAL-AD4630-24FMCZ (24-bit, 2 MSPS)	EVAL-AD4630-24FMCZ	DigiKey/Mouser	\$199
FPGA Board	Digilent ZedBoard (Zynq-7020)	AES-Z7EV-7Z020-G	Avnet/DigiKey	\$495
SD Card	Kuiper Linux pre-configured	EVAL-SD-KUIPERZ	Included with ADC board	—

Important note about the Avisoft microphone: The standard CM24/CMPA version outputs via 5-pole XLR and requires the UltraSoundGate unit. For direct ADC connection, you must order the CM24/CMPA-5V or CM24/CMPA40-5V variant, which provides BNC output, 5V USB power input, and integrated 200V polarization generator. Contact Avisoft Bioacoustics directly to request this variant based on product code 40013.

2.2 Power System Components

Item	Part Number / Model	Supplier	Purpose	Price
LiFePO4 Battery Pack	TalentCell LF4021	Amazon	Main 12V system power	~\$100
DC Barrel Adapter	DC5521-to-DC5525	Amazon/eBay	Battery to ZedBoard connector	\$5-10
Isolated 5V Supply	LT3045-based module or quality USB bank	DigiKey/Amazon	Clean preamp power	\$20-50
Backup: USB-C PD Trigger	HUSB238 + LM2596 buck	Amazon/ AliExpress	Laptop power option	\$15-25

The TalentCell LF4021 uses LiFePO4 chemistry providing 153.6Wh capacity with stable 12-13V output throughout discharge. Most generic 12V battery packs use 3-cell Li-ion that drops to 9V when depleted. The ZedBoard uses a 5.5mm x 2.5mm barrel jack while most battery packs output 5.5mm x 2.1mm, requiring the adapter cable.

2.3 Signal Conditioning Components

A 4th-order Sallen-Key Butterworth anti-aliasing filter with approximately 100 kHz cutoff frequency is required. The EVAL-AD4630-24FMCZ includes only a single-pole RC filter for settling time, which is insufficient for anti-aliasing at 24-bit precision.

Component	Value	Quantity	Part Number Example	Notes
Op-amp	ADA4896-2 or OPA2210	1	ADA4896-2ARMZ	Low noise, high bandwidth
Resistors	10 k Ω 0.1%	4	RN73C2A10KBTDF	Precision thin film
C1 (Stage 1)	220 pF C0G	2	06035A221JAT2A	Low temperature coefficient
C2 (Stage 1)	330 pF C0G	2	06035A331JAT2A	Low temperature coefficient
C3 (Stage 2)	100 pF C0G	2	06035A101JAT2A	Low temperature coefficient
C4 (Stage 2)	680 pF C0G	2	06035A681JAT2A	Low temperature coefficient

This filter design yields f_c of approximately 100 kHz with approximately 70 dB attenuation at 250 kHz (the Nyquist frequency for 500 kSPS sampling), adequate for preserving 24-bit precision.

2.4 Cables and Adapters

Item	Part Number	Supplier	Purpose
Cat6 Ethernet cable 2m	1847-11111-ND	DigiKey	Primary data interface
SMA-SMA cable RG316 1m	415-0029-MM-00012	DigiKey	Analog signal path
BNC-to-SMA adapter F-M	242105	Amphenol/DigiKey	Microphone to filter interface
USB-A to Micro-B cable	380-1427-ND	DigiKey	UART console access

3. Hardware Assembly

This section provides step-by-step instructions for assembling the complete system. Follow these procedures carefully, as incorrect jumper settings can damage components.

3.1 ZedBoard Jumper Configuration

Before applying power for the first time, verify all jumper settings. The most critical setting is VADJ, which configures the FMC I/O bank voltage. An incorrect setting can permanently damage the ADC evaluation board.

Boot Mode Switches (JP7-JP11)

Configure for SD card boot with the following positions:

Jumper	Position	Setting
JP7	Left	GND
JP8	Left	GND
JP9	Right	3V3
JP10	Right	3V3
JP11	Left	GND

Critical Voltage Settings

VADJ SELECT (J18): **Set to 2.5V position.** This jumper configures the FMC I/O bank voltage that interfaces with the ADC board. The EVAL-AD4630-24FMCZ requires 2.5V I/O levels. Setting this incorrectly will damage the ADC board when power is applied.

USB Power Jumpers (JP2, JP3): **Install both jumpers** to enable USB peripheral power for console access during initial setup.

3.2 Mechanical Assembly Sequence

Follow this sequence to assemble the hardware safely:

1. Insert the Kuiper Linux SD card into the J12 slot on the underside of the ZedBoard. The EVAL-AD4630-24FMCZ kit includes a pre-configured SD card with the complete software stack.
2. With power disconnected, align the EVAL-AD4630-24FMCZ FMC connector with the ZedBoard's J1 FMC-LPC connector. Press firmly until the board is fully seated. The connector should engage smoothly without forcing.
3. The ADC board draws all required power through the FMC connector. No separate power cable is needed for the evaluation board.
4. Do not connect the 12V power adapter until all signal connections are complete and verified.

3.3 Signal Chain Connections

The complete signal path from microphone to ADC requires careful attention to connector types and grounding:

Microphone to Filter

5. Connect the CM24/CMPA-5V BNC output to the female side of the BNC-to-SMA adapter.
6. Connect an SMA cable from the adapter to the anti-aliasing filter input.

Filter to ADC

7. Connect the filter output via SMA cable to the EVAL-AD4630-24FMCZ IN_A+ connector.
8. For single-ended operation, connect IN_A- to signal ground. For true differential measurement, leave IN_A- for the negative signal path.

Data and Console Connections

9. Connect the Ethernet cable from ZedBoard J10 directly to the laptop. No network switch is required for point-to-point connection.
10. Connect a USB cable from the laptop to ZedBoard J14 (UART) for serial console access during initial setup. This connection is only needed during configuration and can be removed for field use.

3.4 Power Connections

The power architecture keeps analog and digital domains separated to minimize noise coupling:

ZedBoard Power

Connect the LiFePO4 battery's 12V output through the DC5521-to-DC5525 adapter cable to the ZedBoard barrel jack J20. The ZedBoard accepts 12V at up to 3A, though typical consumption is approximately 10-15W including the ADC board.

Microphone Preamp Power

Critical: Do NOT power the Avisoft preamp from the ZedBoard's USB ports. The ZedBoard's 5V rails carry significant digital switching noise from DDR3 memory (533 MHz), Gigabit Ethernet PHY, and the Zynq SoC's power management. This noise would directly couple into the ultrasonic signal path.

Instead, use a separate isolated 5V supply. Options include a dedicated low-noise LDO module such as the LT3045 or ADM7150 fed from an isolated DC-DC converter, or a high-quality USB power bank with verified low output noise. The preamp draws approximately 500 mA.

3.5 Grounding Strategy

Proper grounding is essential for achieving the full 24-bit measurement quality:

Cable shield grounding: **Ground all cable shields at one end only**, preferably the ADC board end. Grounding both ends creates ground loops that inject noise.

Ethernet isolation: **The Gigabit Ethernet interface includes transformer magnetics** in the RJ-45 connector providing 1500-2500 Vrms galvanic isolation. This completely eliminates ground loops between the measurement system and laptop, which is why Ethernet is strongly preferred over USB.

Analog signal routing: **Keep analog signal cables** physically separated from digital cables (Ethernet, USB) and AC power cables to minimize electromagnetic coupling.

4. Software Installation

The software stack consists of pre-built components from Analog Devices requiring minimal configuration. This section covers both the embedded system setup and host PC installation.

4.1 ZedBoard First Boot

With hardware assembly complete, perform the initial boot and network configuration:

11. Connect the 12V power adapter to ZedBoard barrel jack J20.
12. Wait approximately 30 seconds for boot completion. The red LD0 LED will blink indicating Linux activity.
13. Open a serial terminal on the laptop (115200 baud, 8N1) connected to the USB UART to observe boot messages and access the command line.
14. Login with username 'analog' and password 'analog'.

Network Configuration

Configure a static IP address for reliable connection. Edit the network interfaces file:

```
sudo nano /etc/network/interfaces

# Add the following lines:
auto eth0
iface eth0 inet static
    address 192.168.1.100
    netmask 255.255.255.0
    gateway 192.168.1.1
```

Save the file and restart networking:

```
sudo systemctl restart networking
```

Verify IIO Daemon

The IIO daemon (iiod) provides network access to the ADC. Verify it is running:

```
systemctl status iiod
# Should display: active (running)
```

4.2 Host PC Software Installation

Install the required libraries and tools on your laptop or workstation.

Windows Installation

15. Download the libiio MSI installer from github.com/analogdevicesinc/libiio/releases
16. Run the installer and follow the prompts.

17. Open Command Prompt and verify installation: `iio_info -V`

Linux Installation (Ubuntu/Debian)

```
sudo apt-get update
sudo apt-get install libiio-utils libiio-dev python3-pip
pip3 install pyadi-iio pylibiio numpy scipy matplotlib h5py
```

macOS Installation

```
brew install libiio
pip3 install pyadi-iio pylibiio numpy scipy matplotlib h5py
```

MATLAB Requirements

For direct MATLAB integration using the Analog Devices Precision Toolbox:

- MATLAB R2021b or later
- Communications Toolbox
- DSP System Toolbox
- Signal Processing Toolbox
- Analog Devices Precision Toolbox (provides `adi.AD4630_24.Rx` System Object)

Install the Precision Toolbox from the MATLAB Add-On Explorer or download from the Analog Devices wiki at wiki.analog.com/resources/tools-software/matlab-precision-toolbox.

4.3 System Verification

Before proceeding to data acquisition, verify the complete system is operational.

Test IIO Connectivity

From the host PC, configure its Ethernet adapter for the same subnet (for example, 192.168.1.10 with netmask 255.255.255.0), then test connectivity:

```
iio_info -u ip:192.168.1.100
```

Expected output should show the `ad4630-24` device with two voltage channels, a `sampling_frequency` attribute, and buffer capability. If the device is not found, verify network connectivity with `ping` and check that `iiod` is running on the ZedBoard.

Test Basic Capture

Perform a quick capture test from the command line:

```
iio_readdev -u ip:192.168.1.100 -b 1000 ad4630-24 > test.bin
ls -la test.bin
# Should show a non-zero file size
```

Test Python Acquisition

Run a simple Python capture script to verify the complete toolchain:

```
import adi
import matplotlib.pyplot as plt

adc = adi.ad4630(uri='ip:192.168.1.100', device_name='ad4630-24')
adc.rx_buffer_size = 4096
adc.sample_rate = 500000

data = adc.rx()
print(f'Captured {len(data[0])} samples from {len(data)} channels')

plt.figure()
plt.plot(data[0][:1000])
plt.title('AD4630-24 First Capture')
plt.xlabel('Sample')
plt.ylabel('ADC Code')
plt.show()
```

If this script runs successfully and displays captured data, the system is fully operational and ready for data acquisition workflow development.

5. Data Acquisition Workflow

This section describes the recommended workflow for capturing ultrasonic data from railroad wheel testing, optimized for 10-second burst captures with manual start and stop control.

5.1 Capture Architecture

For your application of manually-triggered 10-second burst captures, the capture-to-DDR-then-transfer architecture is strongly recommended over continuous real-time streaming. This approach provides guaranteed sample continuity and simpler MATLAB integration.

Data Rate Calculations

Understanding the data rates helps configure appropriate buffer sizes:

Parameter	Value	Calculation
Sample Rate	500 kSPS	Selected for 80 kHz bandwidth
Channels	2	Differential pair or dual sensors
Bits per Sample	24 (stored as 32)	Packed in 32-bit container
Data Rate	~3 MB/s	$500k \times 2ch \times 3 \text{ bytes}$
10-Second Capture	~30 MB	Fits easily in 512 MB DDR3
Transfer Time	~0.6 seconds	At 50 MB/s Ethernet throughput

The ZedBoard's 512 MB DDR3 can buffer over 170 seconds of continuous data at these rates, so a 10-second capture uses only about 6 percent of available memory.

Why Capture-Then-Transfer

Architecture	Advantages	Disadvantages
Capture-to-DDR-then-transfer	Guaranteed continuity, simpler code, full bandwidth	~0.6s latency after capture
Continuous real-time streaming	Unlimited duration, real-time monitoring	Network jitter risk, complex sync

For burst capture of hammer strikes where you manually start recording, allow multiple strikes to occur, then manually stop, the capture-then-transfer approach eliminates any risk of dropped samples due to network timing issues.

5.2 MATLAB Direct Acquisition

The Analog Devices Precision Toolbox provides native MATLAB support for the AD4630-24. The `adi.AD4630_24.Rx` System Object handles all IIO communication internally.

Basic Capture Script

```
% MATLAB acquisition using Precision Toolbox
rx = adi.AD4630_24.Rx;
rx.uri = 'ip:192.168.1.100';
rx.SamplesPerFrame = 5000000; % 10 seconds at 500 kSPS
rx.EnabledChannels = [1, 2];

% This call blocks until buffer fills, then transfers to workspace
disp('Recording... perform hammer strikes now');
data = rx();
disp('Capture complete');

% Data is now in MATLAB workspace for your processing pipeline
release(rx);
```

Integration with Your Processing Pipeline

Since you already have burst detection and isolation algorithms coded in MATLAB, the captured data array can be passed directly to your existing functions:

```
% After capture, data is a matrix: samples x channels
ch1 = data(:, 1); % Channel 1 data
ch2 = data(:, 2); % Channel 2 data
fs = 500000;      % Sample rate

% Call your existing burst detection function
bursts = detectBursts(ch1, fs, threshold);

% Process each detected burst
for i = 1:length(bursts)
    segment = ch1(bursts(i).start:bursts(i).end);
    results(i) = analyzeWheelResponse(segment, fs);
end
```

5.3 Alternative: Python Capture with MATLAB Handoff

If the Precision Toolbox proves problematic, an alternative workflow uses Python for reliable hardware communication with file-based handoff to MATLAB:

Python Capture Script

```
# Python capture script (capture_burst.py)
import adi
import numpy as np
from scipy.io import savemat
import time

adc = adi.ad4630(uri='ip:192.168.1.100', device_name='ad4630-24')
adc.rx_buffer_size = 5000000 # 10 seconds at 500 kSPS
adc.sample_rate = 500000
```

```

print('Recording... perform hammer strikes now')
data = adc.rx()
print('Capture complete')

# Save for MATLAB
savemat('capture.mat', {
    'ch0': data[0],
    'ch1': data[1],
    'sample_rate': adc.sample_rate,
    'timestamp': time.time()
})
print(f'Saved {len(data[0])} samples to capture.mat')

```

MATLAB Loading and Analysis

```

% Load captured data
load('capture.mat')

% Create time vector
t = (0:length(ch0)-1) / sample_rate;

% Visualize raw capture
figure
subplot(2,1,1)
plot(t * 1000, ch0)
xlabel('Time (ms)')
ylabel('ADC Code')
title('Channel 0 - Raw Capture')

% Compute spectrum
subplot(2,1,2)
[pxx, f] = pwelch(double(ch0), [], [], [], sample_rate);
plot(f/1000, 10*log10(pxx))
xlabel('Frequency (kHz)')
ylabel('PSD (dB/Hz)')
title('Power Spectral Density')

```

5.4 Calling Python from MATLAB

MATLAB can also call Python functions directly using its built-in Python engine, providing a hybrid approach:

```

% Configure Python environment (one-time setup)
pyenv('Version', '/usr/bin/python3'); % Adjust path as needed

% Import pyadi-iio
adi = py.importlib.import_module('adi');

% Create ADC object
adc = adi.ad4630(uri='ip:192.168.1.100', device_name='ad4630-24');
adc.rx_buffer_size = int32(5000000);

```

```
adc.sample_rate = int32(500000);  
  
% Capture data  
py_data = adc.rx();  
  
% Convert to MATLAB arrays  
ch0 = double(py.numpy.array(py_data{1}));  
ch1 = double(py.numpy.array(py_data{2}));
```


6. Field Deployment

This section provides practical guidance for using the system in field conditions for railroad wheel testing.

6.1 Pre-Deployment Checklist

Complete these items before leaving the laboratory:

- Fully charge the LiFePO4 battery (153Wh provides approximately 10 hours at 15W system draw)
- Verify the isolated 5V preamp supply is charged and operational
- Confirm MATLAB Precision Toolbox recognizes ZedBoard over Ethernet
- Test the complete capture workflow end-to-end in laboratory conditions
- Pack spare cables (Ethernet, SMA, BNC adapter) for field redundancy
- Ensure laptop has sufficient storage for accumulated capture files

6.2 Field Setup Procedure

Follow this sequence at the field site:

18. Position the equipment case near the test location, keeping the laptop in a stable, shaded position if possible.
19. Connect power: LiFePO4 battery to ZedBoard barrel jack via adapter cable; isolated 5V supply to Avisoft preamp USB input.
20. Connect the signal chain: microphone BNC output through BNC-SMA adapter to anti-aliasing filter input; filter output to ADC IN_A+ via SMA cable.
21. Connect Ethernet cable directly from ZedBoard J10 to laptop. No network switch is required.
22. Wait approximately 30 seconds for ZedBoard to complete booting after power is applied.
23. Configure the laptop's Ethernet adapter for a static IP on the same subnet (for example, 192.168.1.10).
24. Verify connectivity by pinging 192.168.1.100 or running `iio_info`.

6.3 Data Collection Workflow

For each test sequence:

25. Position the microphone at the designated measurement point on or near the railroad wheel.
26. In MATLAB, initialize the acquisition object and configure buffer size for your expected capture duration.
27. Execute the capture command. The system will begin buffering data to DDR3.
28. Perform hammer strikes on the wheel while recording is active.

29. When the buffer fills (or you terminate early), data transfers automatically to the MATLAB workspace.
30. Run your burst detection algorithm to identify and isolate individual hammer strike responses.
31. Save the raw capture and processed results with appropriate metadata (wheel ID, location, timestamp).

6.4 Troubleshooting Common Issues

No Network Connection

If the laptop cannot reach the ZedBoard after power-up, verify that both devices are configured for the same subnet. Ensure the Ethernet cable is firmly seated at both ends. Try a different cable to rule out cable damage from field handling.

IIO Device Not Found

If `iio_info` connects but does not show the `ad4630-24` device, the FPGA bitstream may not have loaded correctly. Power cycle the ZedBoard and ensure the SD card is fully seated. Check that `VADJ` is set to 2.5V.

Noisy Measurements

If captured data shows unexpected noise, verify that the preamp is powered from the isolated 5V supply rather than ZedBoard USB. Check that all cable shields are grounded at only one end. Separate analog signal cables from digital and power cables.

Capture Timeout or Incomplete Data

If captures fail to complete or show missing samples, reduce the sample rate or buffer size to verify basic operation first. Ensure no other network traffic is competing for Ethernet bandwidth. Check that the laptop's firewall is not blocking the IIO port (30431).

7. Reference Information

7.1 Complete Bill of Materials Summary

Category	Item	Estimated Cost
Core Hardware	Avisoft CM24/CPMA-5V microphone	€600-800
	EVAL-AD4630-24FMCZ ADC board	\$199
	Digilent ZedBoard	\$495
Power System	TalentCell LF4021 LiFePO4 battery	\$100
	DC barrel adapter cable	\$10
	Isolated 5V preamp supply	\$30
Signal Conditioning	Anti-aliasing filter components	\$50
Cables/Adapters	Ethernet, SMA, BNC adapter, USB	\$50
	TOTAL (excluding microphone)	\$934 + €600-800

7.2 Key Network Settings

Parameter	Value	Notes
ZedBoard IP Address	192.168.1.100	Configure in /etc/network/interfaces
Subnet Mask	255.255.255.0	Standard Class C
IIO Daemon Port	30431	Default iiod port
Host PC IP	192.168.1.x (not 100)	Any address on same subnet
Connection Type	Direct Ethernet	No switch required

7.3 Software Resources

Resource	URL
Analog Devices HDL Repository	github.com/analogdevicesinc/hdl
pyadi-iio Python Library	github.com/analogdevicesinc/pyadi-iio
libiio Library Releases	github.com/analogdevicesinc/libiio/releases
Kuiper Linux Images	wiki.analog.com/resources/tools-software/linux-software/kuiper-linux
MATLAB Precision Toolbox	wiki.analog.com/resources/tools-software/matlab-

	precision-toolbox
AD4630-24 Datasheet	analog.com/AD4630-24
ZedBoard Documentation	digilent.com/reference/programmable-logic/zedboard/start

7.4 Implementation Timeline

Phase	Duration	Activities
Component Procurement	1-2 weeks	Order all components, await delivery
Hardware Assembly	1-2 days	Jumper configuration, mechanical assembly, signal connections
Software Installation	1 day	ZedBoard network setup, host PC libraries, MATLAB toolbox
System Verification	1 day	Connectivity tests, basic captures, noise characterization
Workflow Development	2-3 days	MATLAB integration, burst detection pipeline adaptation
Field Testing	1-2 days	Initial field deployment, procedure refinement
Total	2-4 weeks	From procurement to field-ready system

7.5 Document Revision History

This document consolidates the complete system design including core DAQ architecture, power system design for field deployment, and MATLAB integration workflow. The system leverages mature Analog Devices evaluation hardware and pre-built software to minimize custom development while achieving professional-grade 24-bit ultrasonic measurement quality suitable for railroad wheel defect characterization.

End of Document